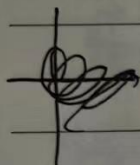


第十三章复习答案

1. (4) T. $(\vec{r}(t) \cdot \vec{r}(t))' = 2\vec{r}(t) \cdot \frac{d\vec{r}}{dt} = 0 \Rightarrow |\vec{r}(t)| = c$

(5) F. $y = x^2 \Rightarrow y' = 2x, y'' = 2 \Rightarrow y'(0) = 0, y''(0) = 2$

$$\kappa = \frac{|y''|}{[1 + (y')^2]^{\frac{3}{2}}} = 2$$



$$x^2 + y^2 = 1 \Rightarrow \begin{cases} x = \cos t \\ y = \sin t \end{cases} \Rightarrow \left| \frac{d\vec{r}}{dt} \right| = 1, |\vec{r}(t)| = 1 \Rightarrow \kappa = 1$$

2. C. $\vec{v}(t) = \left(\frac{2}{t^3}, \frac{3}{t^2}, 2t \right) \quad p(2, -2, 1) \Rightarrow t = 1.$

$$\Rightarrow \vec{v}(t) = (-2, 3, 2)$$

$$\cdot (-1, -2, 2) = 2 - 6 + 4 = 0.$$

3. C. $\vec{v}(t) = (t^{\frac{1}{2}}, -(2-t)^{\frac{1}{2}}, 1) \Rightarrow |\vec{v}(t)| = \sqrt{t + 2 - t + 1} = \sqrt{3}$

$$L = \int_{\frac{1}{4}}^{\frac{1}{2}} |\vec{v}(t)| dt = \frac{\sqrt{3}}{4}$$

4. $\vec{a}(t) = (e^t, t, \sin 2t) \Rightarrow \vec{v}(t) = (e^t, \frac{1}{2}t^2, \frac{-\cos 2t}{2}) + \vec{c} \Rightarrow \vec{v}(0) = (1, 0, \frac{1}{2}) + \vec{c}$

$$\vec{v}(0) = (1, 0, 0) \Rightarrow \vec{c} = (0, 0, \frac{1}{2})$$

$$= (1, 0, 0)$$

$$\Rightarrow \vec{v}(t) = (e^t, \frac{1}{2}t^2, \frac{1 - \cos 2t}{2})$$

$$\Rightarrow \vec{r}(t) = (e^t, \frac{1}{6}t^3, \frac{t}{2} - \frac{\sin 2t}{4}) + \vec{c}_1$$

$$\vec{r}(0) = (1, 0, 0) + \vec{c}_1 = (0, 0, 0) \Rightarrow \vec{c}_1 = (-1, 0, 0)$$

$$\Rightarrow \vec{r}(t) = (e^t - 1, \frac{t^3}{6}, \frac{t}{2} - \frac{\sin 2t}{4})$$

$$5. |\vec{r}(t)| = c \Rightarrow \vec{r}(t) \cdot \vec{r}(t) = c^2$$

$$\Rightarrow 2 \vec{r}(t) \cdot \frac{d\vec{r}}{dt} = \underline{\underline{0}}$$

$$6. \vec{r}(t) = (e^t \cos t, e^t \sin t, 2)$$

$$\vec{v}(t) = (e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, 0)$$

$$|\vec{v}(t)| = \sqrt{e^{2t} (\cos t - \sin t)^2 + e^{2t} (\sin t + \cos t)^2} = \sqrt{2e^{2t}} = \sqrt{2} e^t$$

$$\vec{T}(t) = \frac{1}{|\vec{v}(t)|} \vec{v}(t) = \frac{1}{\sqrt{2}} (\cos t - \sin t, \sin t + \cos t, 0)$$

$$\frac{d\vec{T}}{dt} = \frac{1}{\sqrt{2}} (-\sin t - \cos t, \cos t - \sin t, 0)$$

$$\left| \frac{d\vec{T}}{dt} \right| = \sqrt{\frac{1}{2} [(-\sin t - \cos t)^2 + (\cos t - \sin t)^2]} = 1$$

$$\Rightarrow \kappa = \frac{1}{|\vec{v}|} \left| \frac{d\vec{T}}{dt} \right| = \underline{\underline{\frac{1}{\sqrt{2} e^t}}}$$

$$7. y = \sin x \Rightarrow y' = \cos x, y'' = -\sin x$$

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{\frac{3}{2}}} = \frac{|\sin x|}{(1 + \cos^2 x)^{\frac{3}{2}}} = \frac{|\sin x|}{(2 - \sin^2 x)^{\frac{3}{2}}}$$

$$|\sin x| \uparrow \quad \sin^2 x \downarrow \quad \frac{1}{2 - \sin^2 x} \uparrow \Rightarrow \kappa_{\max} \text{ when } |\sin x| = 1$$

$$\Rightarrow \kappa = \frac{1}{(2-1)^{\frac{3}{2}}} = 1$$

8. (1) $\vec{r}(t) = (\sqrt{2}\cos t, \sin t, \sin t)$

$$\vec{v}(t) = (-\sqrt{2}\sin t, \cos t, \cos t) \Rightarrow |\vec{v}(t)| = \sqrt{2\sin^2 t + \cos^2 t} = \sqrt{2}$$

$$\Rightarrow \vec{T} = \left(-\sin t, \frac{\cos t}{\sqrt{2}}, \frac{\cos t}{\sqrt{2}}\right)$$

$$\frac{d\vec{T}}{dt} = \left(-\cos t, -\frac{\sin t}{\sqrt{2}}, -\frac{\sin t}{\sqrt{2}}\right)$$

$$\left|\frac{d\vec{T}}{dt}\right| = 1$$

$$\Rightarrow \vec{N} = \frac{d\vec{T}/dt}{|d\vec{T}/dt|} = \left(-\cos t, -\frac{\sin t}{\sqrt{2}}, -\frac{\sin t}{\sqrt{2}}\right)$$

$$(2) \kappa = \left|\frac{d\vec{T}}{dt}\right| \frac{1}{|\vec{v}(t)|} = \frac{1}{\sqrt{2}}$$

9. $\vec{r}(t) = (t+3, 5-t^2)$

$$\vec{v}(t) = (1, -2t), |\vec{v}(t)| = \sqrt{1+4t^2} = \sqrt{1+t^2}$$

$$\vec{T} = \frac{1}{|\vec{v}(t)|} \vec{v}(t) = \left(\frac{1}{\sqrt{1+t^2}}, -\frac{t}{\sqrt{1+t^2}}\right) \checkmark$$

$$\frac{d\vec{T}}{dt} = \left(\frac{-t}{(1+t^2)^{\frac{3}{2}}}, -\frac{1}{\sqrt{1+t^2}} + \frac{t^2}{(1+t^2)^{\frac{3}{2}}}\right) = \left(\frac{-t}{(1+t^2)^{\frac{3}{2}}}, \frac{1}{(1+t^2)^{\frac{3}{2}}}\right)$$

$$\left|\frac{d\vec{T}}{dt}\right| = \frac{\sqrt{t^2+1}}{\sqrt{(1+t^2)^3}} = \frac{1}{1+t^2}$$

$$\Rightarrow \vec{N} = \frac{d\vec{T}/dt}{|d\vec{T}/dt|} = \left(\frac{-t}{\sqrt{1+t^2}}, \frac{1}{\sqrt{1+t^2}}\right) \checkmark$$

$$\kappa = \left|\frac{d\vec{T}}{dt}\right| \frac{1}{|\vec{v}(t)|} = \frac{1}{2(1+t^2)^{\frac{3}{2}}} \checkmark$$

$$10. \vec{r}(t) = (f(t), g(t), h(t))$$

$$f'(t) = \cos(t), \quad g'(t) = -\cos t + t \sin t, \quad h'(t) = \sum_{n=1}^{\infty} t^{n-1} = \frac{1}{1-t}$$

$$\Rightarrow f'(0) = 1, \quad g'(0) = -1, \quad h'(0) = 1$$

$$\Rightarrow \vec{r}'(0) = (1, -1, 1)$$

$$11. \vec{r}(t) = (2 \sin t, -2 \cos t, 5t) \Rightarrow \vec{v}(t) = (2 \cos t, 2 \sin t, 5)$$

$$|\vec{v}(t)| = 13$$

$$(0, -2, 0) \Rightarrow t = 0$$

$$\Rightarrow L = \int_0^t |\vec{v}(t)| dt = 26\pi \Rightarrow 13t = 26\pi \Rightarrow t = 2\pi$$

$$P(0, -2, 10\pi)$$

$$12. y = x^2 \Rightarrow y' = 2x, \quad y'' = 2 \Rightarrow y'(1) = 2, \quad y'' = 2$$

$$\Rightarrow \kappa = \frac{|y''|}{[1 + (y')^2]^{3/2}} = \frac{2}{(1+4)^{3/2}} = \frac{2}{5\sqrt{5}}$$

$$\text{B } r = \frac{1}{\kappa} = \frac{5\sqrt{5}}{2}$$

$$\text{the normal line: } y - 1 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{x}{2} + \frac{3}{2}$$

$$\begin{cases} y = -\frac{x_0}{2} + \frac{3}{2} \\ (x_0 - 1)^2 + (y_0 - 1)^2 = \frac{25}{4} \end{cases}$$

$$(x_0 - 1)^2 + (y_0 - 1)^2 = \frac{25}{4}$$

$$\Rightarrow (x + 4)^2 + (y - \frac{7}{2})^2 = \frac{125}{4}$$

$$\Rightarrow \begin{cases} x_0 = 6 \\ y_0 = \frac{3}{2} \end{cases} \quad \text{or} \quad \begin{cases} x_0 = -4 \\ y_0 = \frac{7}{2} \end{cases}$$

$$7. P(1, 0, -2), Q(2, -1, 3) \Rightarrow \vec{PQ} = (1, -1, 5), \vec{PO} = (1, 0, -2)$$

$$\Rightarrow \text{the direction of } \vec{PO} \text{ is } \frac{\vec{PQ}}{|\vec{PQ}|} = \frac{1}{\sqrt{27}}(1, -1, 5) = \frac{1}{3\sqrt{3}}(1, -1, 5)$$

$$\Rightarrow \vec{v_0} = |\vec{v_0}| \cdot \frac{\vec{PQ}}{|\vec{PQ}|} = \frac{1}{\sqrt{3}}(1, -1, 5)$$

$$\vec{a_0} = (1, 2, 1)$$

$$\Rightarrow \vec{v_0} = (t, 2t, t) + \vec{c_1} \Rightarrow \vec{v_0} = \vec{c_1} = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{5}{\sqrt{3}}\right)$$

$$\Rightarrow \vec{v_0} = \left(t + \frac{1}{\sqrt{3}}, 2t - \frac{1}{\sqrt{3}}, t + \frac{5}{\sqrt{3}}\right)$$

$$\vec{H_0} = \left(\frac{1}{2}t^2 + \frac{t}{\sqrt{3}}, t^2 - \frac{t}{\sqrt{3}}, \frac{1}{2}t^2 + \frac{5}{\sqrt{3}}t\right) + \vec{c_2}$$

$$\Rightarrow \vec{H_0} = \vec{c_2} = (1, 0, -2)$$

$$\Rightarrow \vec{H_0} = \left(\frac{1}{2}t^2 + \frac{t}{\sqrt{3}} + 1, t^2 - \frac{t}{\sqrt{3}}, \frac{1}{2}t^2 + \frac{5}{\sqrt{3}}t - 2\right).$$